

XRayVision AI — Project Documentation

Final Year Project documentation set for **XRayVision AI**, a full-stack medical diagnostic platform.

Field	Value
Project title	Web-Based AI Chatbot for Automated X-Ray Fracture Detection
Institution	Minhaj University Lahore — School of Software Engineering
Programme	BSSE, 8th Semester — Final Year Project
Students	Muhammad Ali Raza (2022F-MUL-BSSWE-017) · Hamza Afzal (2022F-MUL-BSSWE-027)
Supervisor	Maam Misbah — Lecturer, School of Software Engineering
Product version	2.1.0
Live application	https://x-ray-vision-board.vercel.app

The Documents

#	Document	What it answers	Size
0	Handover note	Start here. What you have, the five things to do, the three things not to do, and the honest status.	1 page
★	Explained Simply	No jargon. What the project is, every technology explained one by one, how the AI works, and likely viva questions with answers. Start here if you are new to the project.	Plain language
1	Software Requirements Specification	<i>What must the system do?</i> Scope, users, 60+ numbered functional requirements, use case model, non-functional targets, constraints.	IEEE 830 format
2	Software Design Document	<i>How is it built?</i> Architecture, ERD, DFDs (levels 0–2), class diagram, sequence diagrams, state diagrams, algorithms, security design, deployment.	IEEE 1016 format
3	Test Documentation	<i>How was it verified?</i> Test plan, environment, test data, 93 test case specifications, traceability, defect log.	IEEE 829 format
4	User Manual	<i>How do I use it?</i> Step-by-step guide to every feature, with screenshots from the live deployment, troubleshooting and FAQ.	End-user guide
5	Thesis	The bound submission document. Title page, certificate, declaration, abstract, eight chapters, IEEE references and four appendices — drawing on documents 1–4.	FYP thesis

■ For submission, use [XRayVision AI Thesis.docx](#) — the thesis formatted for binding: A4, Times New Roman 12 pt, 1.5 line spacing, 1.5-inch binding margin, roman-numeral front matter, arabic

body pagination, and an automatic Table of Contents field. See [Printing and Binding](#).

 **Every document also exists as a ready-to-open .html file in this folder** with all diagrams already rendered — no editor extension required, and it prints straight to PDF. See [Viewing the Diagrams](#).

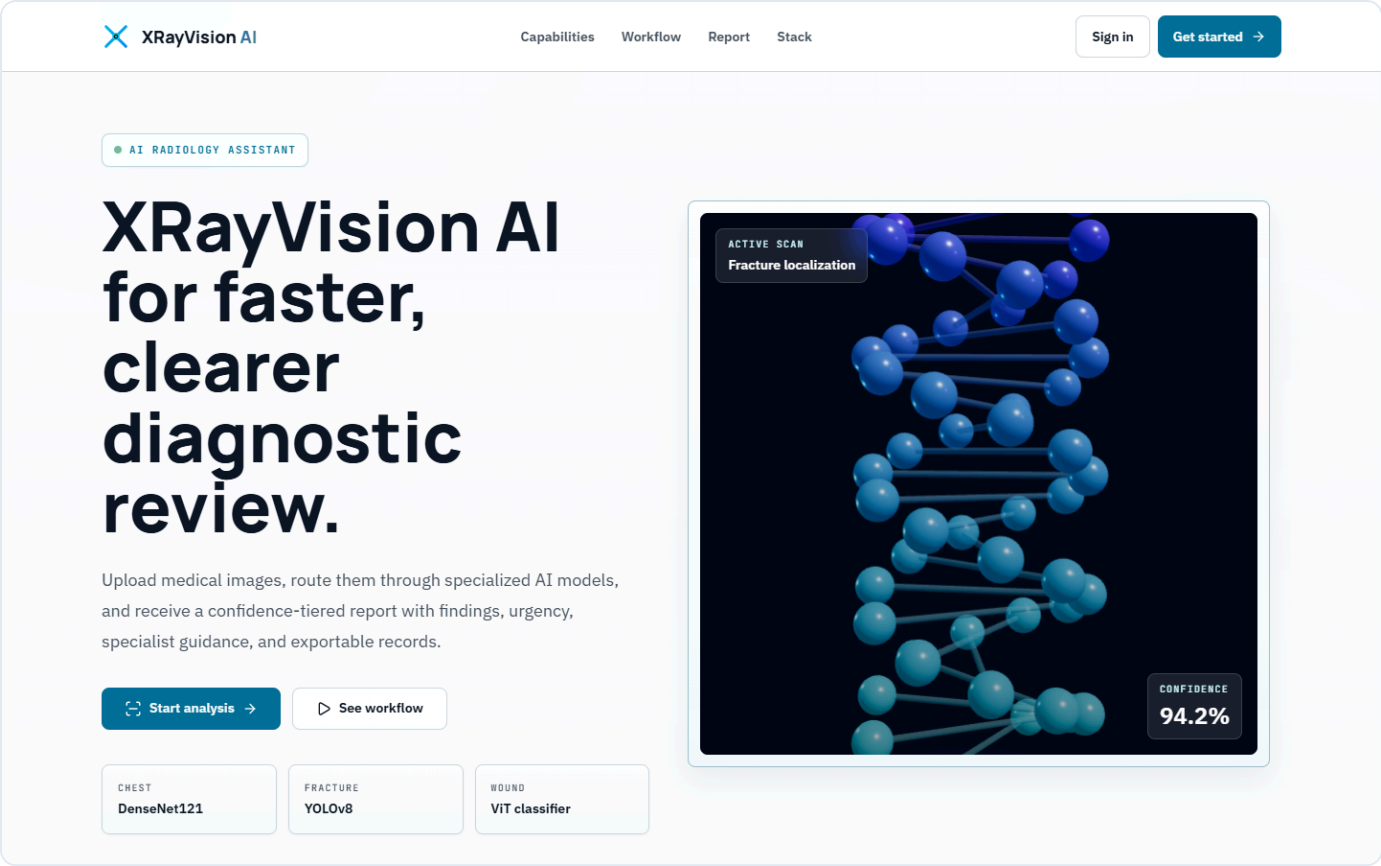
Supporting material also in the repository:

Document	Location
Enterprise Technical Documentation (27 pages, PDF)	../XRayVision_AI_Documentation.pdf
Project overview and quick start	../README.md
Backend setup and model reference	../backend/README.md
Security policy	../SECURITY.md
Contribution guidelines	../CONTRIBUTING.md
Code of conduct	../CODE_OF_CONDUCT.md
Database schema	../backend/supabase_schema.sql

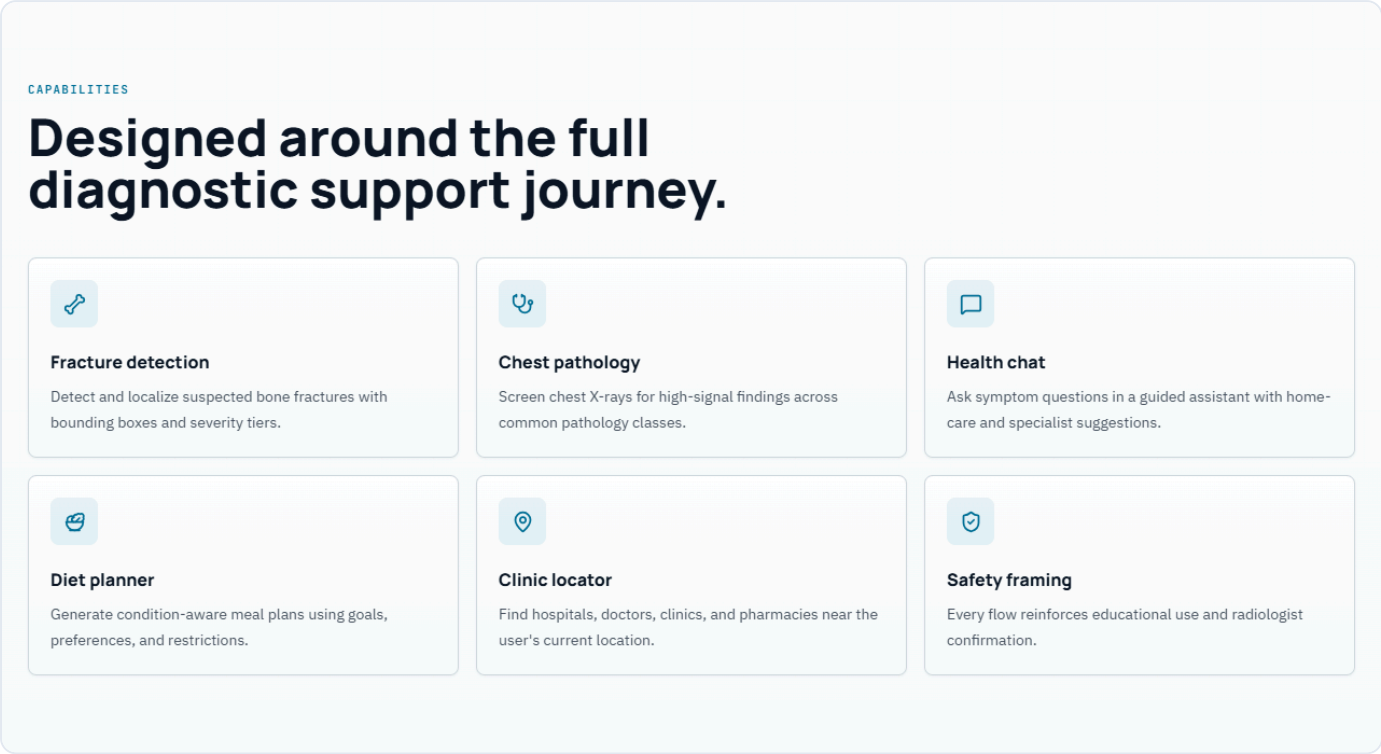
Screenshot Gallery

Every screenshot below was captured from the **live deployment** at <https://x-ray-vision-board.vercel.app> using an automated Chromium session signed in to the demo account. The source files are in [screenshots/](#).

Public pages



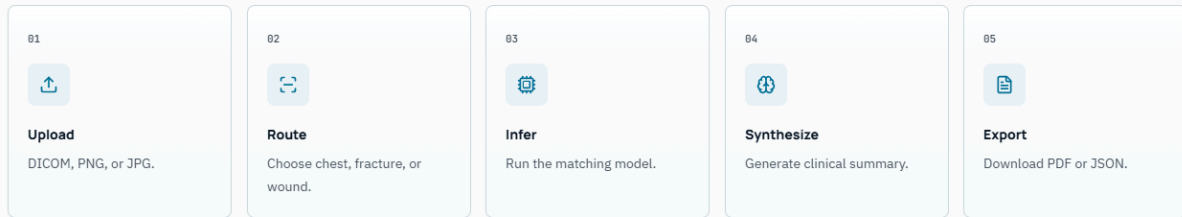
Landing page — 01a-landing-hero.png



Capabilities section — 01b-landing-capabilities.png

WORKFLOW

A transparent path from image to report.



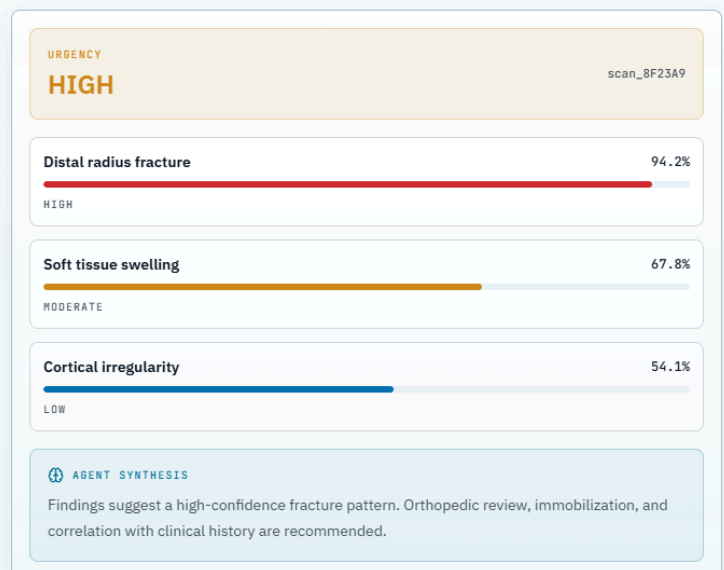
Workflow section — 01c-landing-workflow.png

REPORT EXPERIENCE

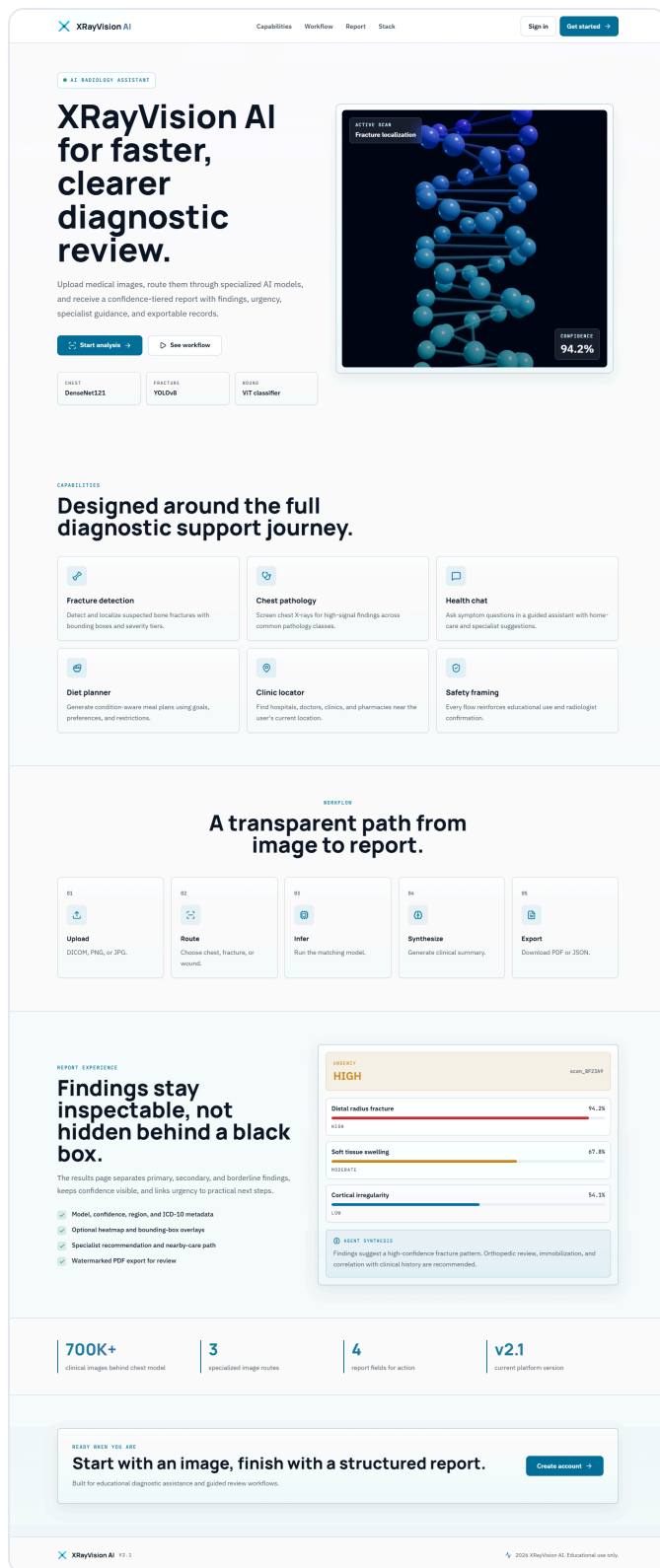
Findings stay inspectable, not hidden behind a black box.

The results page separates primary, secondary, and borderline findings, keeps confidence visible, and links urgency to practical next steps.

- ✓ Model, confidence, region, and ICD-10 metadata
- ✓ Optional heatmap and bounding-box overlays
- ✓ Specialist recommendation and nearby-care path
- ✓ Watermarked PDF export for review



Report-experience section — 01d-landing-report.png



Landing page, full length — 01-landing.png

Authentication

 **XRayVision AI**

SECURE CLINICAL WORKSPACE

Diagnostic assistance with a clean chain of evidence.

Sign in to upload scans, review confidence-tiered findings, export reports, and continue care planning from one focused workspace.

**Multi-model analysis**

Chest, fracture, and wound routes stay separated for cleaner output.

**Educational guardrails**

Every report keeps clinical review and safety disclaimers visible.

**Private records**

Scan history and chat sessions are scoped to the authenticated user.

Educational use only. Not a substitute for a licensed radiologist.

ACCOUNT ACCESS

Welcome back

Sign in to access your diagnostic workspace

EMAIL

PASSWORD

☐ Keep me signed in [Forgot password?](#)

Sign In

or

 Try demo account

New here? [Create an account](#)

Sign-in, including the one-click demo account — 02-login.png

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ACCOUNT ACCESS

Create your account

Free for educational use — no credit card required

FULL NAME

EMAIL

PASSWORD

CONFIRM PASSWORD

I AM A

Medical Student

Healthcare Professional

Researcher

Other

☐ I acknowledge that XRayVision AI is for educational and diagnostic assistance only, and is not a substitute for a licensed radiologist.

Create Account

Already have an account? [Sign in](#)

Registration — 03-register.png

XRayVision AI — Minhaj University Lahore

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SECURE CLINICAL WORKSPACE

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Multi-model analysis

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ACCOUNT ACCESS

Reset your password

We'll send a secure reset link to your inbox.

EMAIL

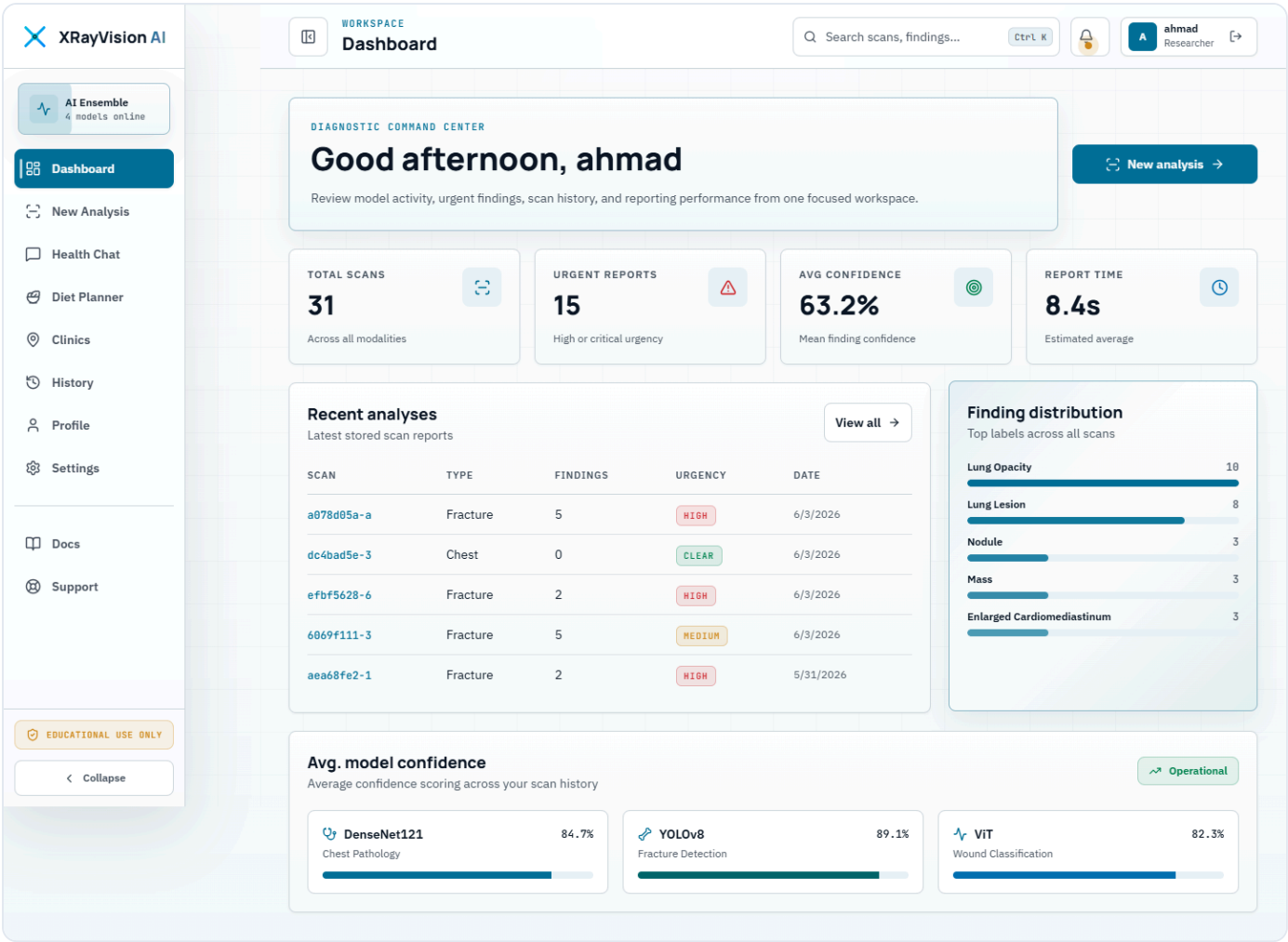
you@hospital.org

Send reset link

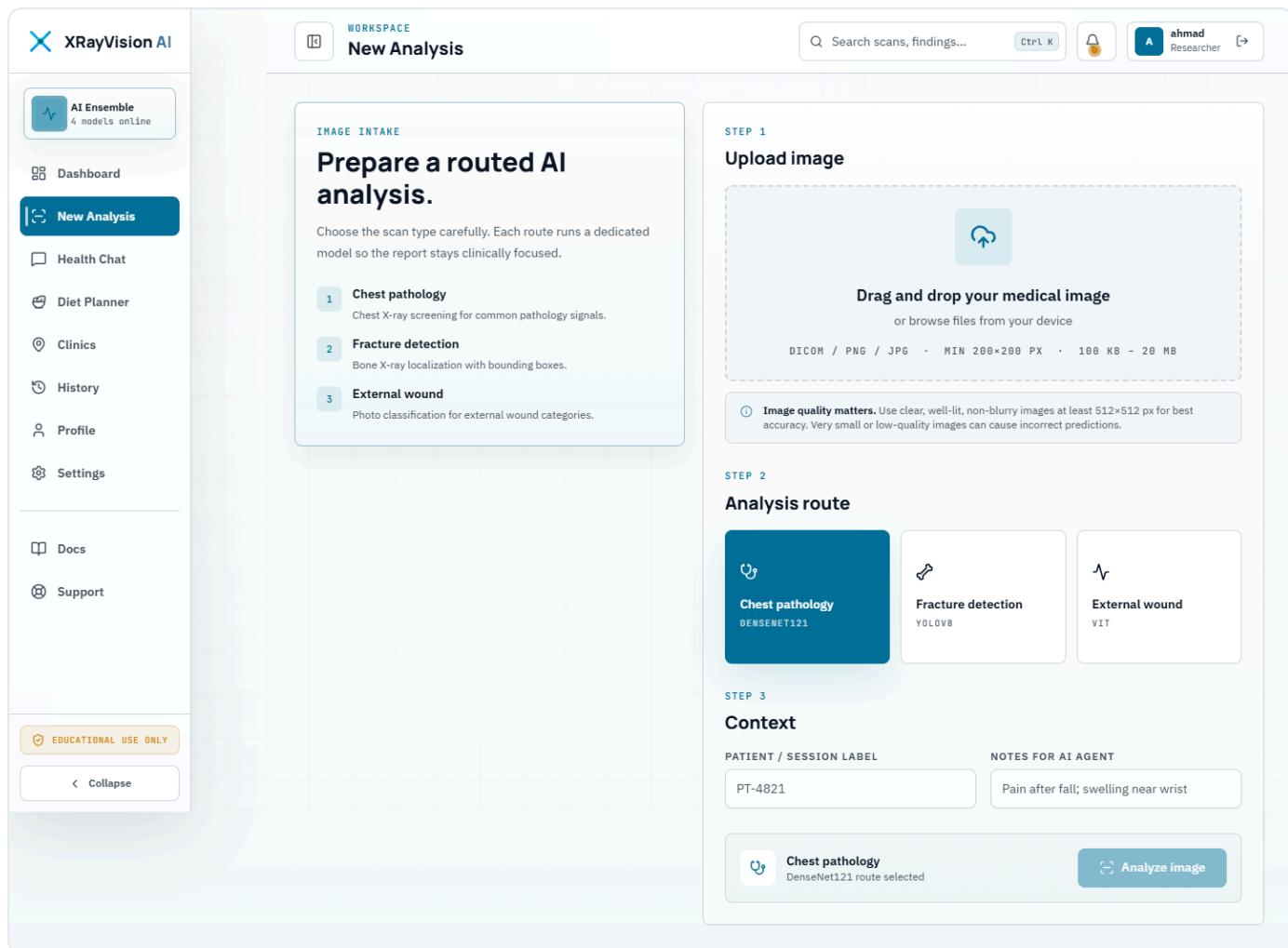
← Back to sign in

Password reset — 04-forgot-password.png

The application



Dashboard — per-user analytics, recent analyses, finding distribution and per-model confidence — 05-dashboards.png



New Analysis — the three-step upload wizard — 06-analyze-upload.png

AI Ensemble

4 models online

Dashboard

New Analysis

Health Chat

Diet Planner

Clinics

History

Profile

Settings

Docs

Support

EDUCATIONAL USE ONLY

Collapse

WORKSPACE

Diagnostic Results

Q Search scans, findings...

Ctrl K

ahmad Researcher

Diagnostic Image

FRACTURE

2026-06-03T10:42:45.156859+00:00

AI Findings

Heatmap

Labels

Q

Q

↺

URGENCY

HIGH

a078d05a-a3ca-4e18-b47e-db76167371f2

One or more findings have confidence below 60% — radiologist review recommended.

Primary Findings

Fracture suspected

HIGH

96.7%

FRACTURECLASSIFIER · FULL IMAGE · ICD-10 S02-S92

Secondary Findings

50-65% confidence

Vase

52.7%

LOW

3 borderline findings below 50% — click to expand

GLM AGENT ANALYSIS

High-confidence fracture detection (96.7%) with high severity indicates significant bone injury requiring immediate attention. The 'Vase' findings appear to be false positives with low confidence (52.7% and below) and low severity, likely representing imaging artifacts. Note: This analysis is for educational purposes only and should not replace evaluation by a qualified radiologist.

Immediate Actions

1 Immediate orthopedic consultation for suspected fracture

2 Additional imaging (CT or MRI) for better characterization of the fracture

3 Pain management and immobilization of the affected area

4 Follow-up imaging to assess fracture healing

Recommended specialist: Orthopedic Surgeon

Find nearby →

Confidence Summary

MODEL	FINDING	CONFIDENCE	SEVERITY
FractureClassifier	Fracture suspected	96.7%	high
YOLOv8	Vase	52.7%	low
YOLOv8	Vase	48.3%	low
YOLOv8	Vase	23.8%	low
YOLOv8	Vase	16.8%	low

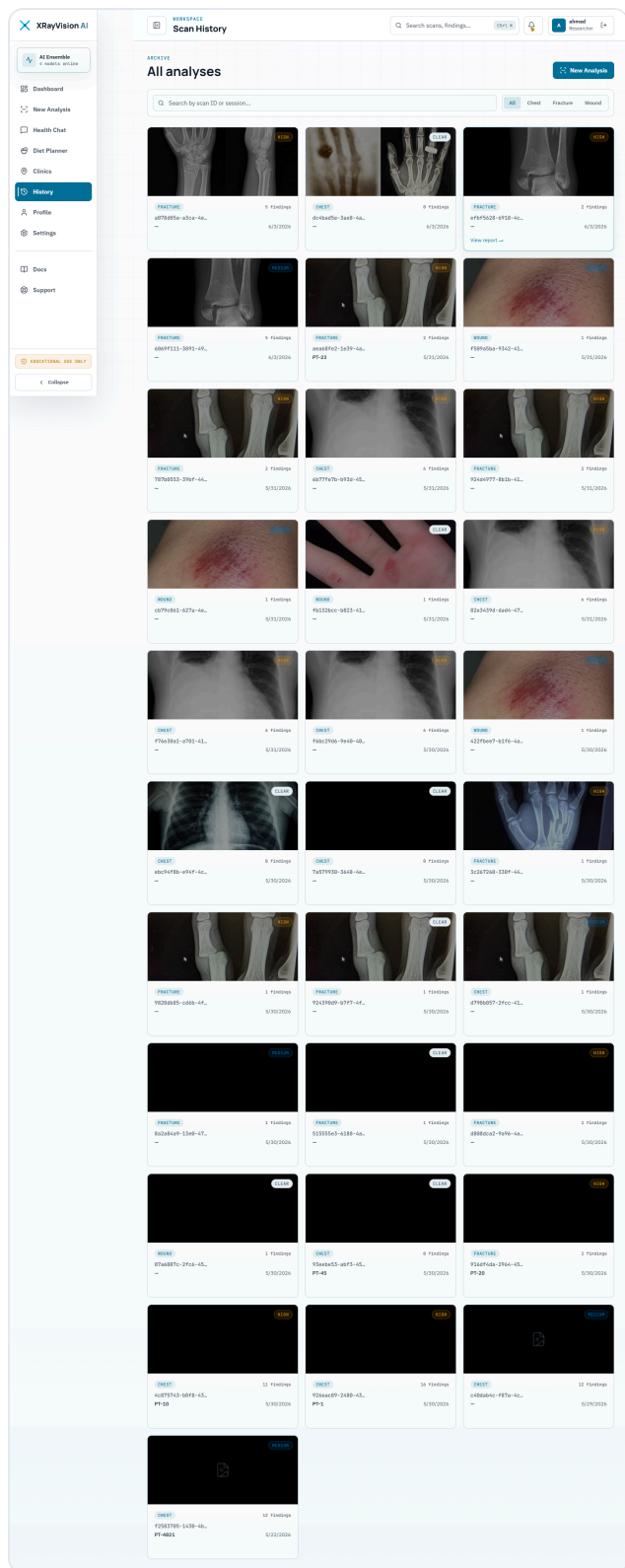
Download PDF Report

Export JSON

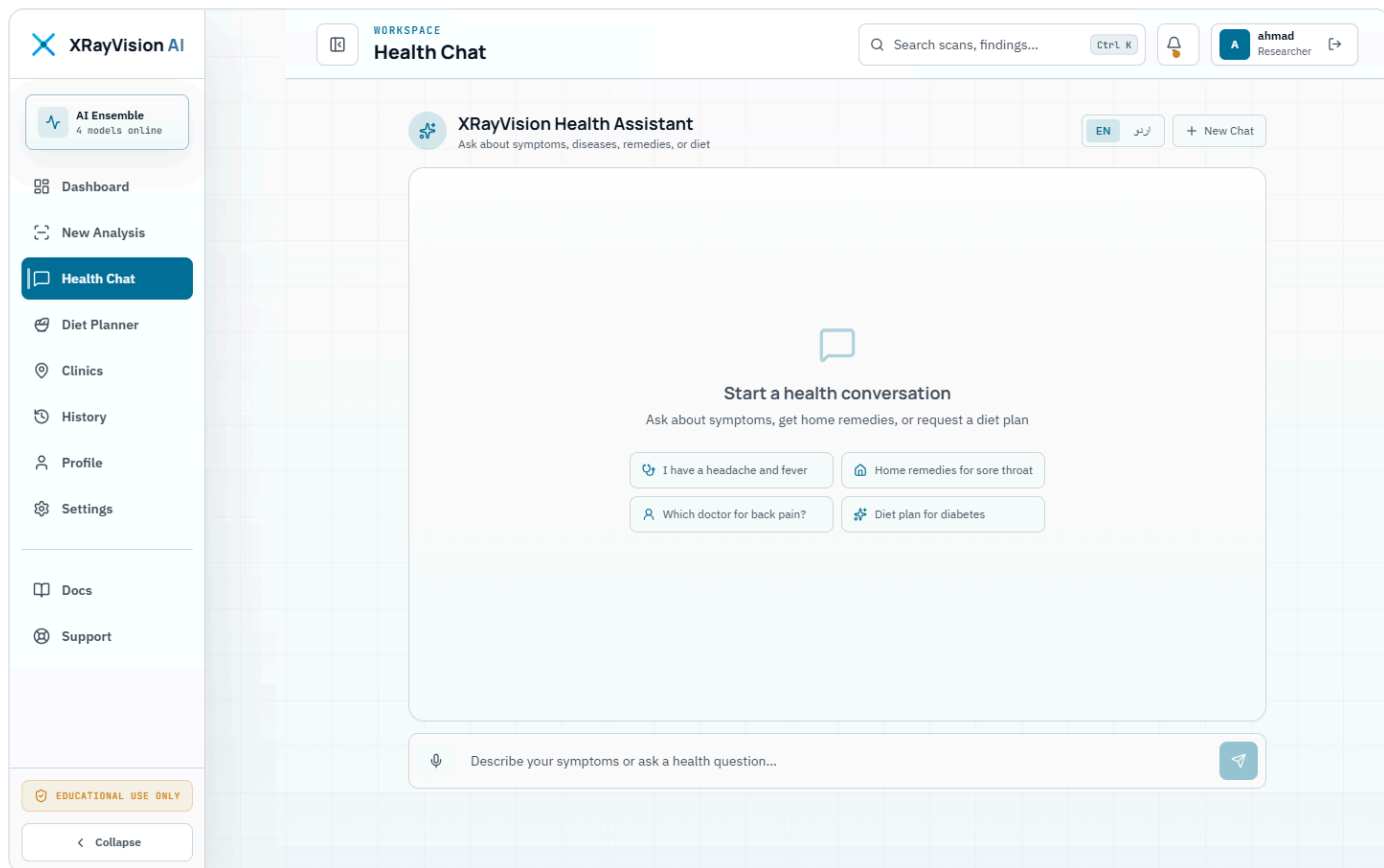
Back to history →

Educational use only. XRayVision AI is not a licensed medical device. All outputs are AI-generated estimations intended to support, not replace, the clinical judgment of a qualified radiologist.

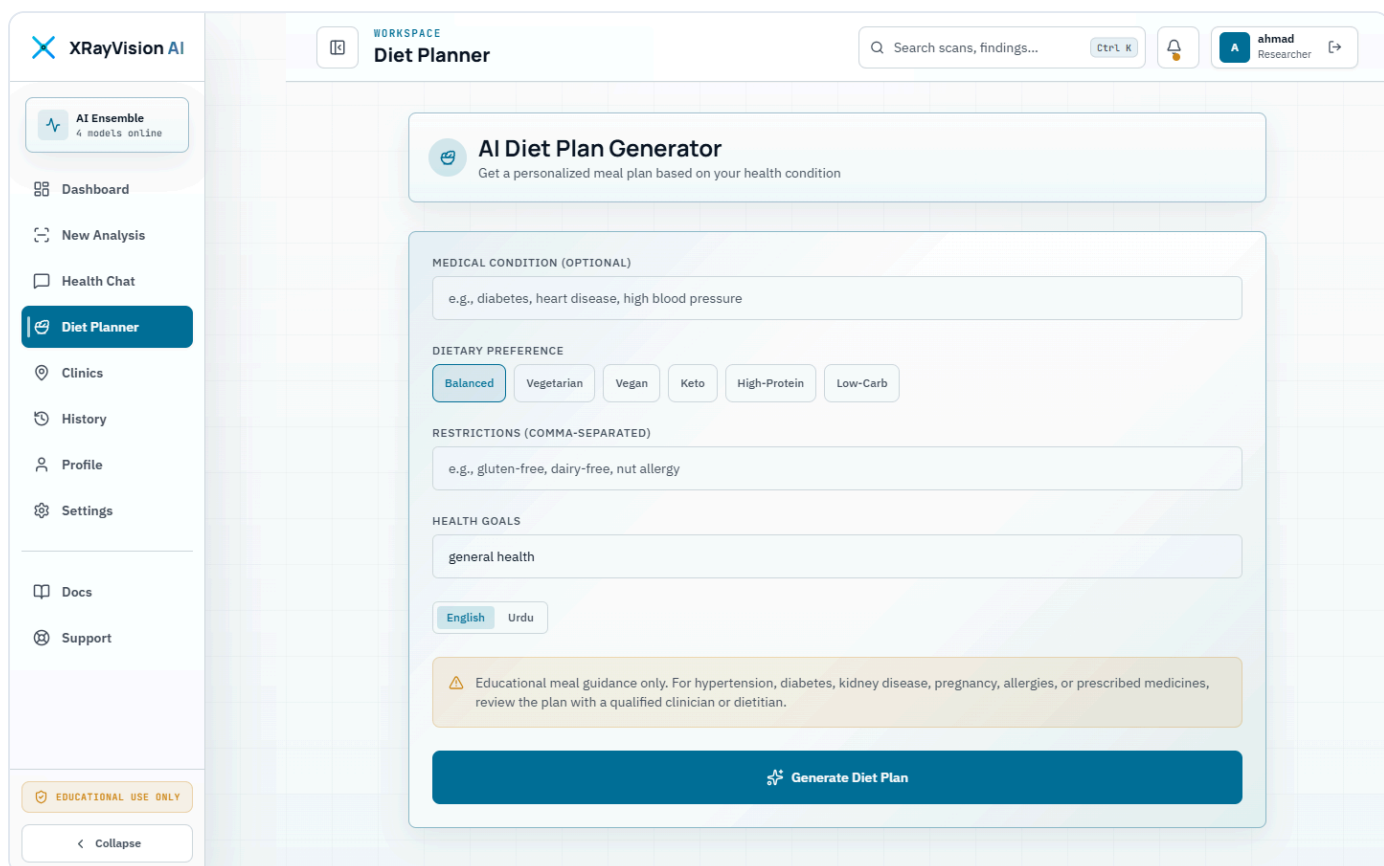
Diagnostic report — bounding-box overlays, tiered findings, low-confidence banner, LLM synthesis, recommended actions and the full confidence table — 13-results.png



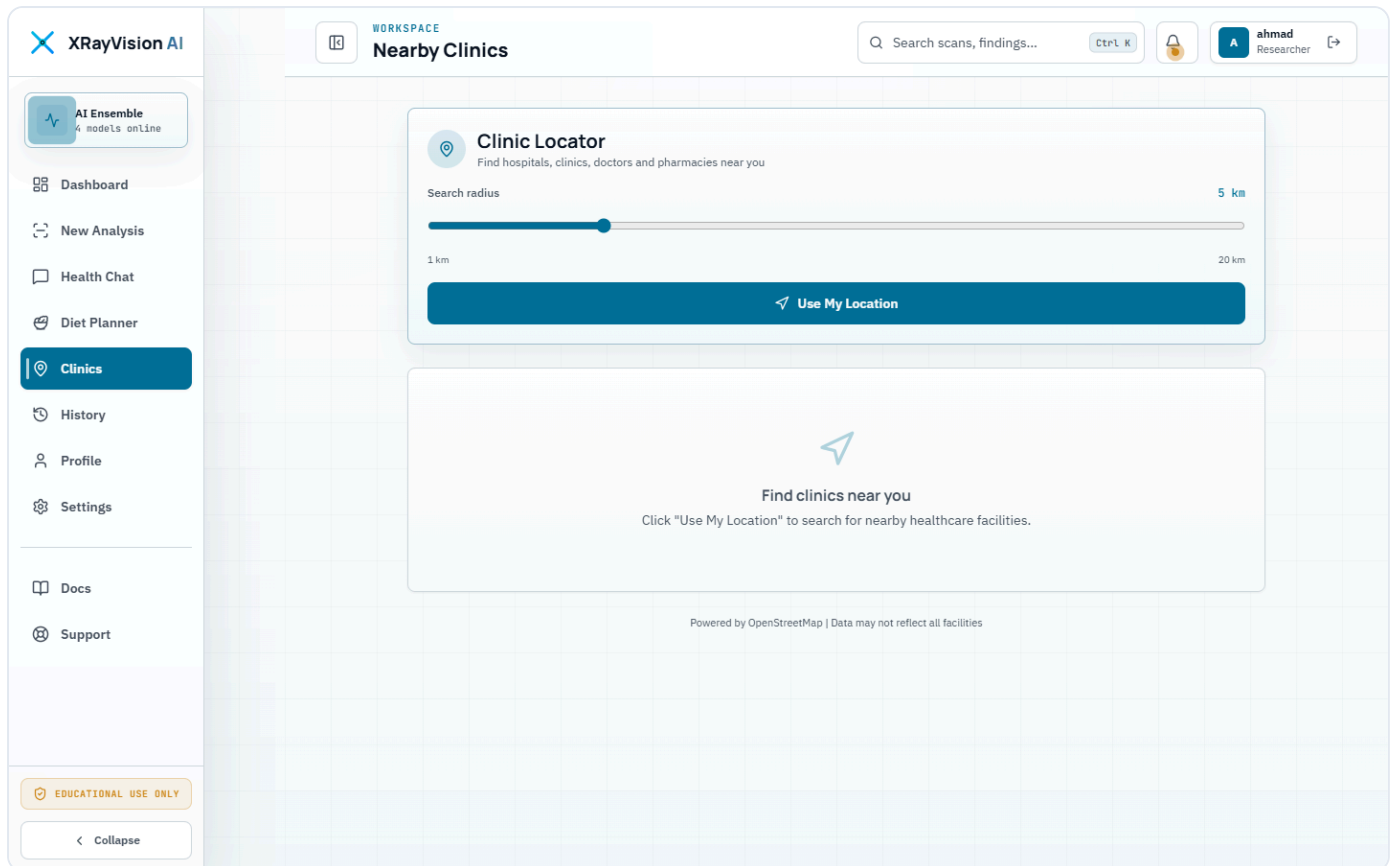
Scan history — 07-history.png



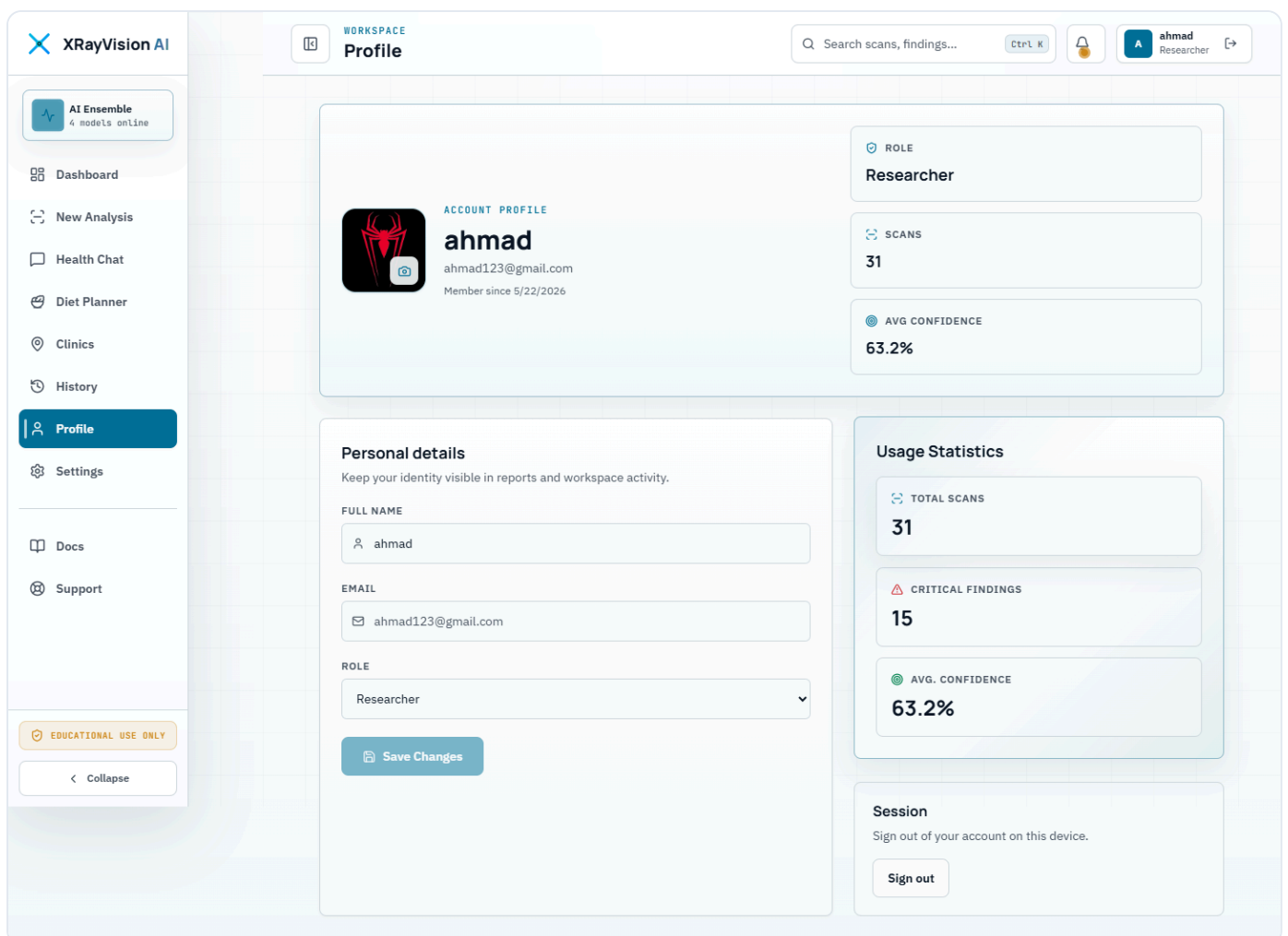
Health chatbot — English/Urdu toggle with voice input — 08-chat.png

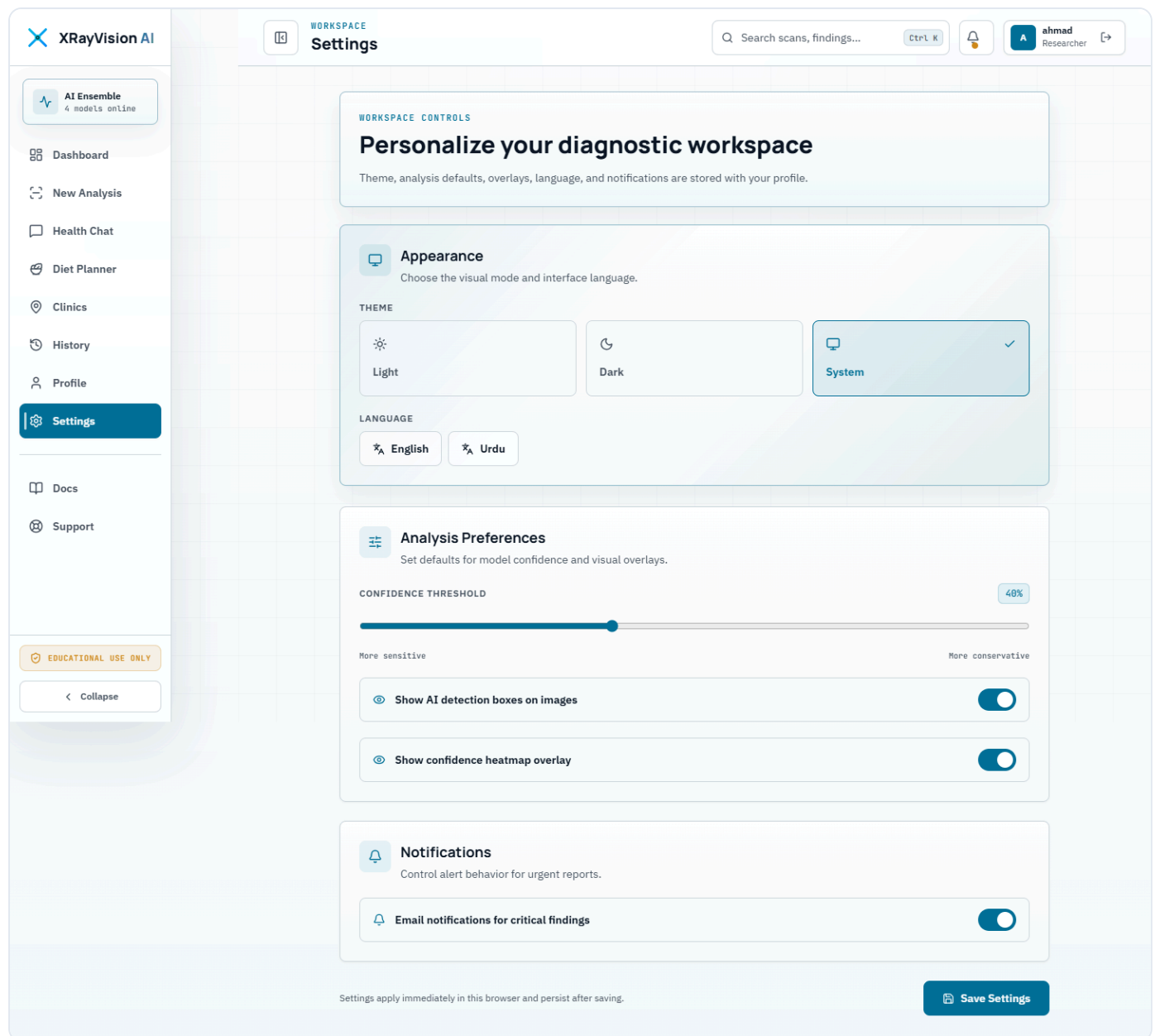


Diet planner — 09-diet.png

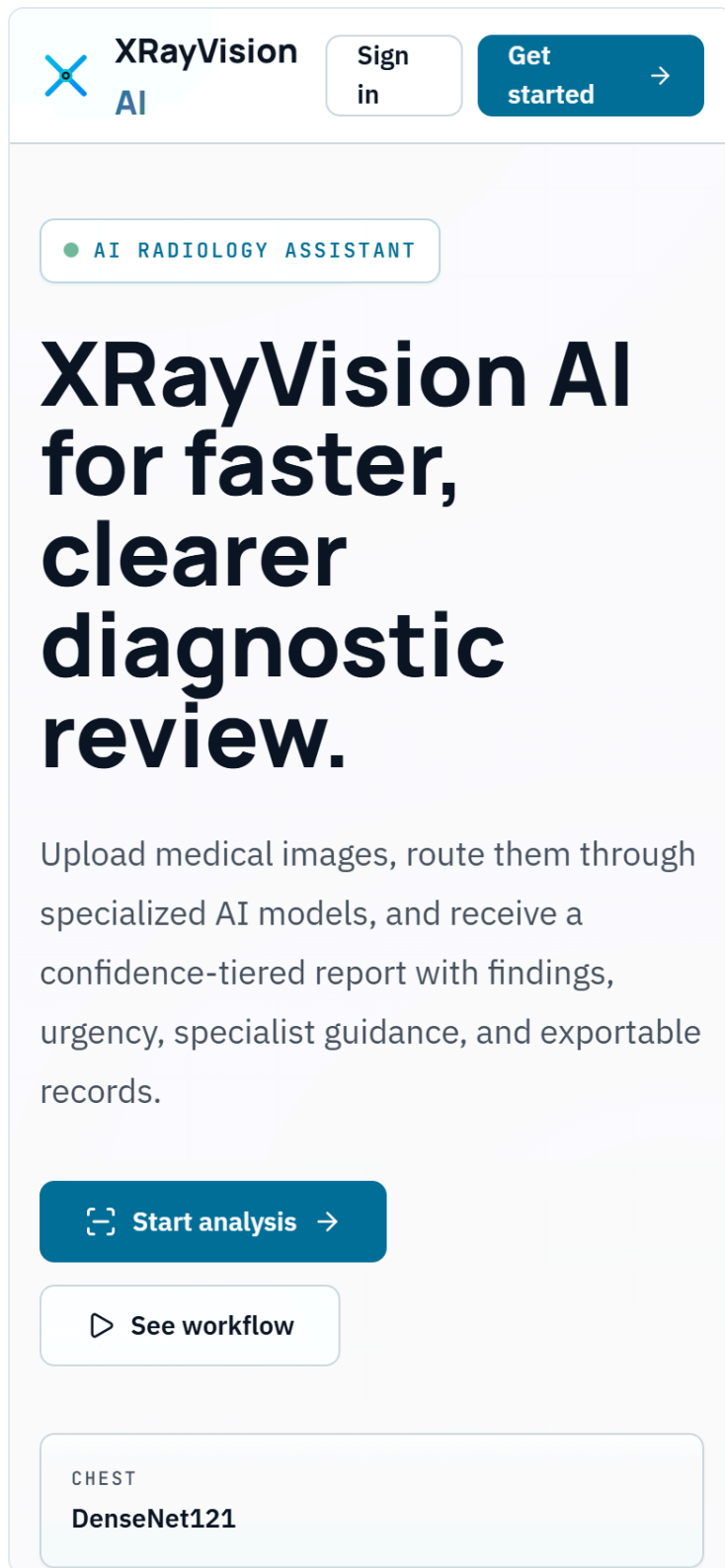


Clinic locator — radius control and OpenStreetMap attribution — 10-clinics.png





Responsive



Mobile layout at 390 × 844 — 14-landing-mobile.png

Also in [screenshots/](#) but not shown above: 01e-Landing-trust.png (the statistics strip, too wide and short to reproduce usefully at page scale).

Reading Order

For the evaluation committee

1. [SRS §1–2](#) — what the system is and who it is for
2. [SDD §2](#) — the architecture
3. [User Manual §6–7](#) — the core feature in action
4. [Test Documentation §16](#) — verification status

For a developer joining the project

1. [../README.md](#) — set up and run it
2. [SDD §5](#) — where the code lives
3. [SDD §6](#) — the API contract
4. [SDD §12](#) — why it is built this way

For an end user Go straight to the [User Manual](#).

Viewing the Diagrams

The diagrams are written in **Mermaid** inside the Markdown files. Depending on how you open the docs, you may need to do nothing — or you may see nothing at all. Here is what works:

✓ Easiest — open the ready-made HTML

Every document has been pre-rendered to a **self-contained HTML file** with all diagrams baked in as SVG. **Just double-click it.** No extension, no internet connection, no build step.

Open this	For
README.html	This index
01-SRS.html	Requirements specification
02-SDD.html	Design document — all 18 diagrams rendered
03-Test-Cases.html	Test documentation
04-User-Manual.html	User manual with screenshots
05-Thesis.html	Thesis, browser edition
06-Simple-Guide.html	Plain-language guide

■ Or just take the PDFs

Every document is also pre-printed to A4 PDF in [pdf/](#) , with page numbers in the footer. Nothing to install, nothing to render — email them, upload them, print them. The index PDF includes the full screenshot gallery.

File	Pages	Contents
00-START-HERE-Handover.pdf	3	Handover note — read first
01-Project-Explained-Simply.pdf	14	Plain-language guide — the stack and the project, no jargon
02-Documentation-Index.pdf	21	This index, including the screenshot gallery
03-Software-Requirements-Specification.pdf	25	Requirements specification
04-Software-Design-Document.pdf	32	Design document
05-Test-Documentation.pdf	45	Test documentation
06-User-Manual.pdf	27	User manual
07-Thesis-browser-edition.pdf	79	Thesis, browser edition (not the submission copy)

⚠ **Which thesis file do I submit?** 05-Thesis-browser-edition.pdf is convenient for reading and emailing, but it is **not** the submission copy — it uses the web stylesheet and has no paginated contents table. **The bound copy comes from [XRayVision_AI_Thesis.docx](#)** : open it in Word, press **F9** on the contents field, then *File* → *Save as* → *PDF*. That version carries Times New Roman, 1.5 spacing, the binding margin and real page numbers in the Table of Contents.

You can also print any `.html` yourself with **Ctrl + P** → **Save as PDF** — the stylesheet has print rules that keep tables, diagrams and screenshots from being split across pages.

Printing and Binding

The submission copy is [XRayVision_AI_Thesis.docx](#) (about 11 MB, with all 22 diagrams and 13 screenshots embedded).

Formatting already applied

Setting	Value
Page size	A4
Margins	1.5 in left (binding edge), 1 in top / right / bottom
Body font	Times New Roman, 12 pt

Setting	Value
Line spacing	1.5
Front matter pagination	Lower-case roman numerals (i, ii, iii ...)
Body pagination	Arabic, restarting at 1 from Chapter 1
Headings	Times New Roman bold, outline levels set for the contents field
Tables	27, full content width, header rows shaded and repeated
Figures	35, centred with italic captions

Before you print — two steps in Word

1. **Build the contents table.** Open the file, click the Table of Contents placeholder, then press **F9** (or right-click → *Update Field* → *Update entire table*). Word fills in the entries and their page numbers. Repeat this after any edit that changes pagination.
2. **Fill in the signature blocks** on the Certificate of Approval and Declaration pages, and sign them.

Then check against your department's template. The formatting above follows the common Pakistani university thesis convention, but departments differ on details — required certificate wording, logo placement, font choice, and whether an abstract word limit applies. Compare against the template your department issues and adjust; everything is plain Word formatting, so nothing here is locked.

To regenerate the .docx after editing. The Word file is generated from [05-Thesis.md](#), so edit the Markdown and rebuild rather than editing both. If you edit the .docx directly, treat it as the new master and stop rebuilding, or your changes will be overwritten. Build instructions are in [tools/README.md](#).

Rebuilding the Documents

The Markdown files are the single source of truth; the .html and .docx files are generated. Three scripts in [tools/](#) regenerate them:

```
cd docs/tools
pip install playwright && python -m playwright install chromium
npm install

python build_html.py           # Markdown -> HTML + diagrams/*.svg
python render_diagrams_png.py  # SVG -> print-quality PNG
node build_thesis_docx.js      # Thesis Markdown -> Word
```

See [tools/README.md](#) for prerequisites, the authoring markers used in the thesis, and how to change the Word formatting.

Viewing the Markdown source instead

Where you open the .md file	Do diagrams render?
GitHub / GitLab (web)	✓ Yes, automatically
VS Code — built-in preview	✗ No. VS Code cannot render Mermaid on its own. Install the free extension Markdown Preview Mermaid Support (<code>bierner.markdown-mermaid</code>), then reopen the preview.
Obsidian, Typora	✓ Yes
Plain text editor	✗ You will see the diagram source code

If your diagrams are blank in VS Code, that is expected — it is the editor, not the document. Either install the extension above or just open the matching .html file.

Individual diagram files

Each diagram is also saved separately as an SVG in [diagrams/](#) — 22 files, named after their source document and position (e.g. `02-SDD-diagram-03.svg` is the ERD). Use these to paste a single diagram into a thesis chapter, a slide deck or a poster. SVG is vector, so it stays sharp at any print size.

Diagram inventory

Diagram	Type	Location
Product context	Graph	SRS §2.1
Product functions	Mindmap	SRS §2.2
Use case diagram	Graph	SRS §5.1
Analysis activity diagram	Flowchart	SRS §5.3
Layered architecture	Graph	SDD §2.1
Component diagram	Graph	SDD §2.2
Entity relationship diagram	ER diagram	SDD §3.1
RLS policy map	Graph	SDD §3.5
DFD Level 0 (context)	Graph	SDD §4.1
DFD Level 1	Graph	SDD §4.2
DFD Level 2 (analysis)	Graph	SDD §4.3

Diagram	Type	Location
Domain class diagram	Class diagram	SDD §5.3
Registration/login sequence	Sequence	SDD §7.1
Image analysis sequence	Sequence	SDD §7.2
Chat sequence	Sequence	SDD §7.3
PDF export sequence	Sequence	SDD §7.4
Scan lifecycle	State diagram	SDD §7.5
Auth session lifecycle	State diagram	SDD §7.6
Auth/authorisation flow	Graph	SDD §9.1
Degradation decision tree	Graph	SDD §10
Deployment diagram	Graph	SDD §11.1
Startup sequence	Sequence	SDD §11.2

22 diagrams across the two design documents.

Status

Document	Status
SRS	✅ Complete — 60+ functional requirements, 12 use cases, full NFR set
SDD	✅ Complete — 22 diagrams, algorithms, design rationale
Test Documentation	⚠️ Specifications complete (93 cases). 17 verified against the live deployment; 75 pending execution; 1 defect open. See the notice at the top of the document.
User Manual	✅ Complete — every feature documented with live screenshots
Thesis	✅ Complete — 8 chapters, 25 IEEE references, 4 appendices; Word edition ready for binding

Open items before final submission

1. **Update the Table of Contents field** in the `.docx` (select it and press F9) and sign the certificate and declaration pages.
2. **Execute the pending test cases** and fill in the Actual Result columns — see [§16.1](#). Then update Chapter 6 of the thesis, which currently reports 17 of 93 executed.
3. ~~Close DEF-01~~ — **done**. The root `.env` has been untracked (`git rm --cached .env`); the local file is untouched.

4. ~~Close DEF-02~~ — **done.** `backend/README.md` now states the `xray-images` bucket is public, matching the implementation.
 5. **Fix DEF-03** — root cause found: the bundled `backend/models/fracture_yolov8.pt` is correct (`Fracture` / `Not_Fracture`), but the deployed backend was loading the generic COCO checkpoint `yolov8n.pt` (which contains `vase`). Correct `YOLO_WEIGHTS_PATH` on the deployed Space, and validate the loaded model's class vocabulary at startup instead of trusting its filename.
 6. **Check the thesis against your department's template** — certificate wording, logo placement and font requirements vary by department.
-

Medical Disclaimer

XRaiVision AI is an educational tool. It is NOT a certified medical device. All outputs are AI-generated estimates intended to support, not replace, the clinical judgement of a qualified radiologist. Do not use this software for diagnosis or treatment decisions.

XRaiVision AI — Project Documentation — Minhaj University Lahore